

Non-singular Fixed-time Terminal Sliding Mode Control For A Small Size Unmanned Helicopter

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Abstract. This paper designs disturbance observers for both the position and attitude loops in order to address the problem of disturbances in small-scale unmanned helicopters. In order to overcome the singularity problem introduced by Euler angle representation and the ambiguity problem introduced by quaternion attitude representation, a fixed-time terminal sliding mode controller is developed, using a geometric definition of error. The Lyapunov approach is used to find the greatest stable time. Furthermore, the sliding mode controller's design successfully avoids singularity issues and lowers chattering. Through simulation, the algorithm's efficacy was confirmed.

Keywords: Small size unmanned helicopter, NFTSM, Position and attitude control, Disturbance observer, Geometry control

1 Introduction

Unmanned aerial vehicles (UAVs) have become more and more in demand in the military and civilian sectors in recent years. UAVs have been trending toward miniaturization at the same time [1]. tiny size unmanned helicopters have features like underactuation [2], nonlinearity [3], strong coupling [4], unmodeled dynamics [5], and susceptibility to external disturbances [6], compared to tiny quadcopter UAVs. These elements contribute to complicated dynamic features and a highly nonlinear and linked system model.

The work on variable structure control and sliding mode control is introduced in reference [7]. By including non-linear functions into the sliding hyperplane design, terminal sliding mode control techniques enable the tracking error on the sliding surface to converge to zero in a finite amount of time [8]. A fast terminal sliding mode control (FTSMC) algorithm is presented in [9], which enhances the speed of terminal sliding mode control and offers stability assurance for manipulating a robotic arm.

[10] suggests a non-singular terminal sliding mode controller (NTSMC) for a nonlinear second-order system in order to solve the singularity issue. In the face of problems, this controller maintains normal operation while resolving singularities. In the publication [11], a NTSMC strategy based on integrals is developed to address the quadcopter control problem in the face of external disturbances.

Furthermore, a disturbance observer is made to notice disruptions and adjust for them, which reduces chattering and solves singularity problems.

There is singularity and ambiguity [12] in the attitude representations based on Euler angles and quaternions, respectively. However, because of its smoothness and boundedness, the attitude representation based on rotation matrices can avoid these problems and is more suited for flight control [13]. Geometric control is another name for the rotation matrix attitude representation-based control technique.

The main contributions of this paper are: (1)Disturbance observers are built for the small unmanned helicopter's attitude and position loops. (2)The tracking error is defined in terms of the rotation matrix representation. (3)The dynamic model of the small size unmanned helicopter is addressed by designing a NFTSMC with rapid convergence properties, and its effectiveness is demonstrated through the utilization of Lyapunov function. (4)The proposed algorithm is validated through simulation, demonstrating its effectiveness.

2 Dynamics of the System

The body fixed reference frame is defined as $\{B\} = \{O_B, \vec{x}_B, \vec{y}_B, \vec{z}_B\}$, and the inertia frame as $\{I\} = \{O_I, \vec{x}_I, \vec{y}_I, \vec{z}_I\}$. The center O_B is situated at the unmanned helicopter's center of gravity. Refer to [14], when considering external disturbances, the dynamic model of the small size unmanned helicopter is represented as shown in Equation (1).

$$\begin{cases} \dot{p} = V, & \dot{V} = -\frac{1}{m}RT_m + mge_3 + d \\ \dot{R} = RS(\omega), & J\dot{\omega} = -\omega \times (J\omega) + \tau + \tau_d \end{cases} \quad (1)$$

where R represents the rotation matrix from the body fixed frame to the inertia frame, m represents the mass of the helicopter, g represents the acceleration caused by gravity, and $d(t)$ represents the unknown external disturbances acting on the position loop. The vectors $p(t)$ and $V(t)$ represent the position and velocity of the helicopter in the body fixed reference frame. The unit vector in z_B of the body fixed frame is represented by the notation $e_3 = [0 \ 0 \ 1]^T$. The primary lift produced by the main rotor is represented by the vector $T_m(t) \in \mathbb{R}$, whereas $\tau(t) = [a_s(t) \ b_s(t) \ T_t(t)]$ [15]. J denotes the matrix of inertia. The angular velocity in the body frame $\{B\}$ is denoted by $\omega(t)$. The disturbance in the attitude loop is denoted by τ_d . The skew-symmetric matrix $S(\omega)$ is made up of the vector $\omega = [\omega_1 \ \omega_2 \ \omega_3]$.

Assumption: The disturbance variation is slow, which can be represented by

$$\dot{d} \approx 0 \quad (2)$$

$$\dot{\tau}_d \approx 0 \quad (3)$$

3 Attitude Controller Design

3.1 Attitude Disturbance Observer Design

The following is a representation of the attitude loop model for the little unmanned helicopter

$$J\dot{\omega} = -\omega \times (J\omega) + \tau + \tau_d \quad (4)$$

To mitigate the impact of disturbances on the attitude loop, the following auxiliary function is designed first

$$\dot{z}_o = -L_o(z_o + P_1) + L_o(\omega \times (J\omega) - \tau) \quad (5)$$

where

$$P_1 = L_{a1}\omega, \quad L_o = L_{a1}J^{-1} \quad (6)$$

where L_{a1} is a constant. According to [16], the disturbance observer is designed

$$\hat{\tau}_d = z_o + P_1 \quad (7)$$

Proof: The observer's error is described as follows:

$$\tilde{\tau}_d = \tau_d - \hat{\tau}_d \quad (8)$$

Substituting (3) for the first-order time derivative of (8) results in

$$\begin{aligned} \dot{\tilde{\tau}}_d &= L_o(z_o + P_1) - L_o(\tau_d - J\dot{\omega}) - \dot{P}_1 \\ &= -L_{a1}J^{-1}\tilde{\tau} \end{aligned} \quad (9)$$

Considering the following Lyapunov candidate function

$$V_1 = \tilde{\tau}_d^T \tilde{\tau}_d \quad (10)$$

Compute the first-order time derivative of (10), and substituting (9) (6) into it

$$\begin{aligned} \dot{V}_1 &= 2\tilde{\tau}_d^T \dot{\tilde{\tau}}_d \\ &= -2L_{a1}\tilde{\tau}_d^T J^{-1}\tilde{\tau}_d \leq 0 \end{aligned} \quad (11)$$

Hence, the estimation error can converge within a finite time.

3.2 Attitude Loop Controller Design

Replace the variable τ_d in the dynamical model with $\hat{\tau}_d$, (4) can be rewritten as

$$J\dot{\omega} = -\omega \times (J\omega) + \tau + \hat{\tau}_d \quad (12)$$

The following definitions were made for the attitude error vector $e_R(t) \in \mathbb{R}^3$, angular velocity error vector $e_\omega(t) \in \mathbb{R}^3$, and attitude error function $\Psi(R_d^T R)(t) \in \mathbb{R}$

$$\begin{cases} \Psi(R_d^T, R) = \frac{1}{2} \text{tr}[I_3 - R_d^T R] \\ e_R = \frac{1}{2}(R^T R_d - R_d^T R)^\vee \\ e_\omega = R^T R_d \omega_d - \omega \end{cases} \quad (13)$$

where $\omega_d(t) \in \mathbb{R}^3$ and $R_d(t) \in \mathbb{R}^{3 \times 3}$ represent the required angular velocity and rotation matrix produced by the position loop, respectively, and $\text{tr}(\cdot)$ indicates the matrix trace. Taking the first-order time derivative of the third equation in (13), and substituting equation (12) into the result, obtain

$$J\dot{e}_\omega = -JS(\omega)R^T R_d \omega_d + JR^T R_d \dot{\omega}_d + \omega \times (J\omega) - \tau - \hat{\tau}_d \quad (14)$$

Referring to the reference [17], the following design of a non-singular fixed-time terminal sliding surface is considered

$$s = e_\omega + a_1 e_R^{[b_1]} + a_2 e_R + a_3 f(e_R) \quad (15)$$

where $f(e_R)$ is an auxiliary function

$$f(e_R) = \begin{cases} e_R^{[b_2]}, \|e_R\| \geq \epsilon \\ \beta_1 e_R + \beta_2 e_R^{[2]} + \beta_3 e_R^3, \|e_R\| < \epsilon \end{cases} \quad (16)$$

and the parameters $a_1 > 0$, $a_2 > 0$, $a_3 > 0$, $b_1 > 1$, $b_2 \in (0.5, 1)$, $\epsilon \in (0, 1)$. The introduction of the auxiliary function $f(e_R)$ is used to address the singularity problem of the sliding surface. The above expression utilizes the following operations

$$x^{[y]} = |x|^y \text{sign}(x) \quad (17)$$

To guarantee uninterrupted flow of the control input, function $f(e_R)$ should be continuous and smooth at $e_R \pm \epsilon$. By calculating the temporal derivatives of $f(e_R)$ at the first, second, and ensuring the continuity of $f(e_R)$, $\dot{f}(e_R)$, and $\ddot{f}(e_R)$ at $e_R \pm \epsilon$, the following expressions can be obtained

$$\begin{cases} \beta_1 + \beta_2 \epsilon + \beta_3 \epsilon^2 = \epsilon^{b_2-1} \\ \beta_1 + 2\beta_2 \epsilon + 3\beta_3 \epsilon^2 = b_2 \epsilon^{b_2-1} \\ 2\beta_2 + 6\beta_3 \epsilon = b_2(b_2 - 1) \epsilon^{b_2-2} \end{cases} \quad (18)$$

The expressions for calculating β_1 , β_2 , and β_3 from (18) are as follows

$$\begin{cases} \beta_1 = \frac{b_2^2 - 5b_2 + 6}{2} \epsilon^{b_2-1} \\ \beta_2 = -(b_2^2 - 4b_2 + 3) \epsilon^{b_2-2} \\ \beta_3 = \frac{b_2^2 - 3b_2 + 2}{2} \epsilon^{b_2-3} \end{cases} \quad (19)$$

Based on (4), the following control input is designed

$$\begin{aligned} \tau = & J(-SR^T R_d \omega_d + R^T R_d \dot{\omega}_d) - \hat{\tau}_d + w \times (J\omega) + J((a_1 b_1 \|e_R\|^{b_1-1} + a_2) \dot{e}_R \\ & + a_3 \dot{f} + k_1 s^{c_1} + k_2 s + k_3 s^{c_2} + K\phi(s, \delta)) \end{aligned} \quad (20)$$

$$\phi(s, \delta) = \frac{s}{|s| + \delta} \quad (21)$$

where $K > 0$, $k_1 > 0$, $k_2 > 0$, $k_3 > 0$, $c_1 > 1$, $c_2 \in (0, 1)$.

Remark1: It can be inferred from [17] that for the system

$$\dot{y} = -h_1 y^{[j_1]} - h_2 y - h_3 y^{[j_2]} \quad (22)$$

its maximum settle time

$$T_{max} = \frac{1}{h_2} \left(\frac{1}{j_1 - 1} \ln \left(1 + \frac{h_2}{h_1} \right) + \frac{1}{(1 - j_2)} \ln \left(1 + \frac{h_2}{h_3} \right) \right) \quad (23)$$

where $h_1 > 0$, $h_2 > 0$, $h_3 > 0$, $j_1 > 1$, $j_2 \in (0, 1)$.

Remark2: From [18], it can be deduced that for $\Psi(R_d^T, R) < M < 2$

$$\frac{1}{2} \|e_R\|^2 \leq \Psi(R_d^T, R) \leq \frac{1}{2 - M} \|e_R\|^2 \quad (24)$$

Theorem 1: The FTSMC (20) built for the small size unmanned helicopter (4) can guarantee the convergence of system states within a finite time. The following is the upper bound of the settle time $T(e_R)$.

$$T(e_R) \leq T_{1max} + T_{2max} + T_{3max} \quad (25)$$

where T_{1max} , T_{2max} , T_{3max} will be explained later in the subsequent sections.

Proof: Taking the first-order time derivative of (15), obtains

$$\dot{s} = \dot{e}_\omega + a_1 b_1 |e_R|^{b_1-1} \dot{e}_R + a_2 \dot{e}_R + a_3 \dot{f} \quad (26)$$

Substituting (20) into (14), obtains

$$\dot{e}_\omega = - (a_1 b_1 |e_R|^{b_1-1} + a_2) \dot{e}_R - a_3 \dot{f} - k_1 s^{[c_1]} - k_2 s - k_3 s^{[c_2]} - K\phi(s, \delta) \quad (27)$$

Substituting (27) into (26), obtains

$$\dot{s} = -k_1 s^{[c_1]} - k_2 s - k_3 s^{[c_2]} - K\phi(s, \delta) \quad (28)$$

Considering the following Lyapunov candidate function

$$V_2 = s^2 \quad (29)$$

the obtained first-order time derivative and substituting (28) are as follows

$$\begin{aligned} \dot{V}_2 &= -2K \frac{s^2}{|s| + \delta} - k_1 |s|^{a_1+1} - k_2 s^2 - k_3 |s|^{a_2+1} \\ &\leq -2k_1 V_2^{\frac{a_1+1}{2}} - k_2 V_2 - k_3 V_2^{\frac{a_2+1}{2}} \end{aligned} \quad (30)$$

It follows from *Remark1* that the system state will arrive at $s = 0$, the sliding surface, in T_{1max} .

$$T_{1max} = \frac{1}{k_2} \left(\frac{1}{c_1 + 1} \ln \left(1 + \frac{k_2}{k_1} \right) + \frac{1}{(1 - c_2)} \ln \left(1 + \frac{k_2}{k_3} \right) \right) \quad (31)$$

Due to the variation of the sliding surface defined in $\|e_R\| = \epsilon$, consider $\|e_R\| \geq \epsilon$

$$s = e_\omega + a_1 e_R^{[b_1]} + a_2 e_R + a_3 e_R^{[b_2]} \quad (32)$$

Considering the following Lyapunov candidate function

$$V_3 = \Psi \quad (33)$$

and calculat its first-order time derivative

$$\begin{aligned} \dot{V}_3 &= e_R^T e_\omega \\ &= -a_1 e_R^T e_R^{[b_1]} - a_2 e_R^T e_R - a_3 e_R^T e_R^{[b_2]} \end{aligned} \quad (34)$$

According to *Remark2*

$$\begin{aligned} \dot{V}_3 &= -a_1 e_R^T e_R^{[b_1]} - a_2 e_R^T e_R - a_3 e_R^T e_R^{[b_2]} \\ &\leq -a_1 (2 - M) \Psi^{\frac{b_1+1}{2}} - a_2 (2 - M) \Psi - a_3 (2 - M) \Psi^{\frac{b_2+1}{2}} \end{aligned} \quad (35)$$

Similarly, based on *Remark1*, it can be conclude that

$$T_{2max} = \frac{1}{(2 - M)a_2} \left(\frac{2}{b_1 + 1} \ln \left(1 + \frac{a_2}{a_1} \right) + \frac{2}{(1 - b_2)} \ln \left(1 + \frac{a_2}{a_3} \right) \right) \quad (36)$$

When the tracking error $\|e_R\| < \epsilon$

$$s = e_\omega + a_1 e_R^{[b_1]} + a_2 e_R + a_3 (\beta_1 e_R + \beta_2 e_R^2 + \beta_3 e_R^3) \quad (37)$$

(34) can be rewritten as

$$\begin{aligned} \dot{V}_3 &= -a_1 e_R^T e_R^{[b_1]} - a_2 e_R^T e_R - a_3 \beta_1 e_R^T e_R + a_3 \beta_2 e_R^T e_R^{[2]} + a_3 \beta_3 e_R^T e_R^3 \\ &\leq -a_1 (2 - M) \Psi - (a_2 + a_3 \beta_1) (2 - M) \Psi \end{aligned} \quad (38)$$

According to *Remark1*, it can be deduce that

$$T_{3max} = \frac{1}{(2 - M)(a_2 + a_3 \beta_1)} \left(\frac{2}{b_1 - 1} \ln \left(1 + \frac{a_2 + a_3 \beta_1}{a_1} \right) \right) \quad (39)$$

4 Position Controller Design

4.1 Position Disturbance Observer Design

Design the following auxiliary function firstly

$$\dot{z}_x = -L_x(z_x + P2) + L_x(Re_3 T_m - mge_3) \quad (40)$$

where

$$P_2 = L_{a2}V, \quad L_x = L_{a2}mI^{-1} \quad (41)$$

where L_{a2} is a constant. The following disturbance observer is designed

$$\hat{d} = z_x + P_2 \quad (42)$$

Proof: The position loop disturbance observer proof method is the same as the attitude loop disturbance observer proof.

4.2 Position Loop Controller Design

Based on the disturbance observer, d in (1) can be replaced with $\hat{d}$, then

$$\dot{V} = -\frac{1}{m}RT_m + mge_3 + \hat{d} \quad (43)$$

Define the errors for position $e_x(t)$, velocity $e_v(t)$, and acceleration $e_a(t)$ as

$$e_p = p_d - p, \quad e_v = \dot{p}_d - \dot{p}, \quad e_a = \ddot{p}_d - \ddot{p} \quad (44)$$

Substituting (43) into (44), obtain

$$m\dot{e}_v = m\ddot{p}_d + T_m Re_3 - mge_3 - \hat{d} \quad (45)$$

Design the control input $T_m(t)$ as

$$T_m Re_3 = -k_p e_p - k_v e_v + mge_3 + \hat{d} - m\ddot{p}_d \quad (46)$$

where k_p, k_v are control gain.

Theorem 2: The designed control input $T_m(t)$ guarantees the asymptotic convergence of the position tracking error of the unmanned helicopter to zero.

Proof: Substituting (46) into (45), obtains

$$m\dot{e}_v = -k_p e_p - k_v e_v \quad (47)$$

Considering the following Lyapunov candidate function

$$V_5 = \frac{1}{2}k_p e_p^T e_p + \frac{1}{2m}k_v e_v^T e_v \quad (48)$$

Differentiating (48) and substituting (47) yields

$$\begin{aligned} \dot{V}_5 &= k_p e_p^T e_v - k_p e_v^T e_p - k_v e_v^T e_v \\ &\leq 0 \end{aligned} \quad (49)$$

5 Numerical Simulation

To validate the proposed method, simulation experiments were conducted using MATLAB. A comparison was made with traditional terminal sliding mode control methods.

The given trajectory is $x_d = [0.6\cos(\omega t) \ 0.4\sin(\omega t) \ 0.5t]^T$, $\psi_d = \pi t \text{ rad}$. External disturbance to the system is $d = [0.07\sin(2t) \ 0.06\cos(2t) \ 0.05\sin(2t)]^T$ and $\tau_d = [0.7\sin(2t) \ 0.6\cos(3t) \ 0.5\sin(4t)]^T$. The small unmanned helicopter's control parameters are chosen as $a_1 = 2$, $a_2 = 2$, $a_3 = 4$, $b_1 = \frac{9}{5}$, $b_2 = \frac{7}{9}$, $\epsilon = 0.05$, $k_1 = 25$, $k_2 = 0.1$, $k_3 = 0.6$, $c_1 = \frac{7}{5}$, $c_2 = \frac{5}{7}$, $K = 1.5$, $k_p = 150$, $k_v = 80$, $L_{a1} = 117$, $L_{a2} = 565$, $m=9.8$, $J = \text{diag}([0.18, 0.34, 0.28])$.

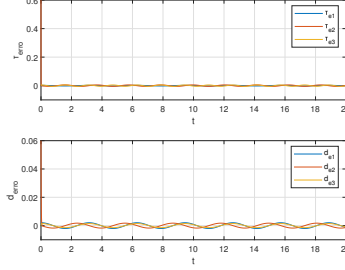


Fig. 1: Observation error of the disturbance observer

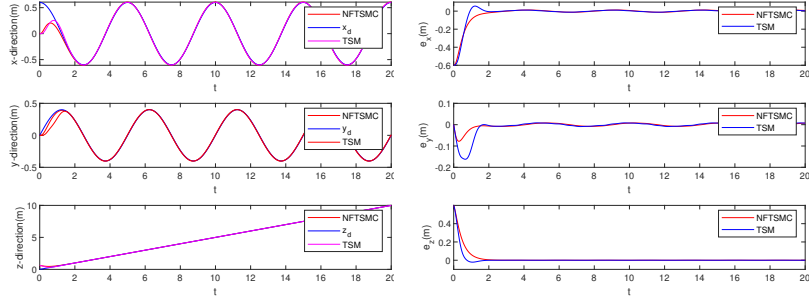


Fig. 2: Comparison of trajectory tracking and tracking error

It is evident from Fig.1 that the disturbance observer constructed in this research has a comparatively minimal observation error. This suggests that the suggested disturbance observer can detect and monitor external disturbances fast and precisely. From Fig.2, it can be observed that both NFTSMC and TSM can

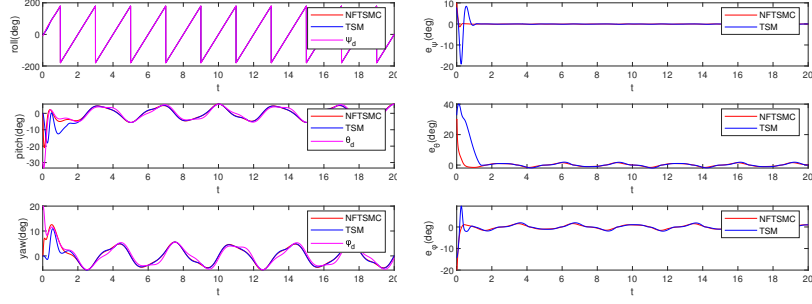


Fig. 3: Comparison of attitude tracking and tracking error

track the desired trajectory quite well. But, compared to the traditional TSMC, the proposed NFTSMC effectively addresses the singularity problem near the equilibrium point. From Fig.3, it can be observed that both NFTSMC and TSMC can track the desired attitude quite well. And the proposed NFTSMC effectively addresses the singularity problem near the equilibrium point. From Fig.4, it can be observed that the traditional TSMC exhibits significant chattering.

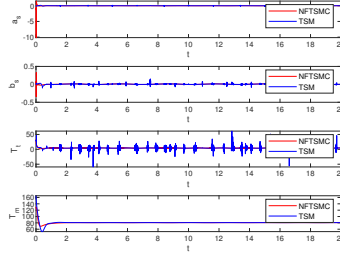


Fig. 4: Comparison of control inputs

6 Conclusion

The findings above demonstrate that the observer can effectively detect external disturbances and adjust the controller accordingly. The proposed control method addresses the singularity and chattering problems of the traditional terminal sliding mode control, while ensuring a good attitude and position tracking effect in the simulation environment. In order to validate the algorithm, future study will take into account setting up a small unmanned helicopter experiment platform.

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